= 1.9 x 10 N

25 B

For a solenoid, the internal B-field is: $B = \mu_0 \dfrac{N}{L} I$

Using Faraday's Law, the induced emf between the rim and center of the disk is:

$$
\begin{aligned}
E &= \frac{d\Phi}{dt} \\
&= B\frac{dA}{dt} \\
&= \mu_0 \frac{N}{L} I \frac{A}{T} \qquad \text{(A radius on the disc will sweep out an area } A \text{ in time } T\text{)} \\
&= \mu_0 \frac{N}{L} IA \frac{\omega}{2\pi} \\
&= 0.16\mu_0 NIA\omega / L
\end{aligned}
$$

Since the emf between the ends of the solenoid is 10 times larger than the induced emf,

$$E_{\text{solenoid}} = 1.6\mu_0 NIA\omega / L$$

26 D